

Energy

WE RELY ON ENERGY TO MAKE OUR WORLD FUNCTION.

Energy is what makes things happen. It is everywhere and in everything, giving objects the ability to move or glow with heat.

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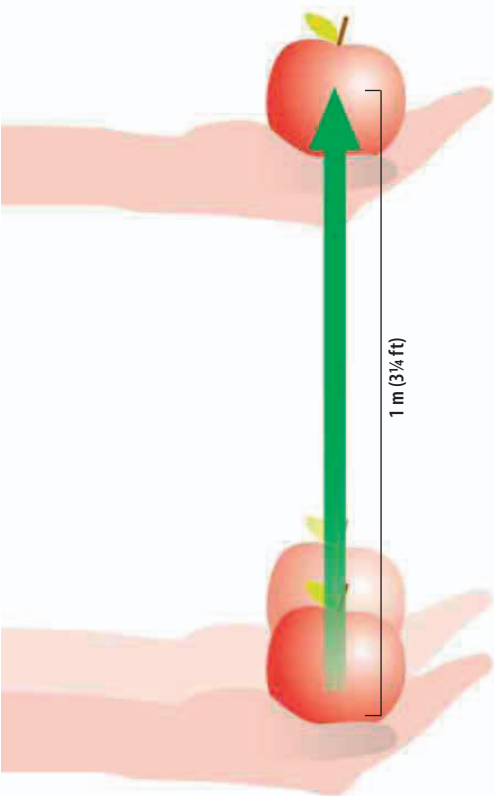
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Measuring energy

Energy can be put to work. To a physicist, the word “work” means the amount of energy involved in moving an object. Work is calculated as the amount of force multiplied by the distance. Since force is measured in newtons (N) and distance in meters (m), such a calculation results in a unit of work called a newton meter (Nm).

▽ One joule of energy

One joule (J) is the amount of energy transferred to an object by a force of 1 N over a distance of 1 m (3¼ ft). This is roughly equivalent to lifting an apple up 1 m (3¼ ft).



Types of energy

Energy can be seen working in many ways. Although they are given different names and appear in a wide range of contexts—from the energy released by an exploding star to the energy in a bouncing ball—all types of energy are closely related, and each one can change into other types (see the opposite page for examples).



◁ Kinetic energy

This is the energy of motion. As an object speeds up, it contains more kinetic energy.



◁ Thermal energy

The air blowing out from a hairdryer is hot because electrical energy is converted into thermal energy.



◁ Electrical energy

This type of energy is carried by an electric current that supplies all kinds of appliances.



◁ Chemical energy

This is the form of energy released when chemical reactions take place, such as burning fuel.



◁ Radiant energy

This is the form of energy carried by light and other types of electromagnetic radiation.



◁ Nuclear energy

This form of energy is released when atoms split apart (fission) or join together (fusion).



◁ Sound energy

This is a type of energy that objects produce when they vibrate in a medium, such as air.



◁ Potential energy

The diver has potential energy due to her or his height above the water, which changes to kinetic energy as the diver falls.